UNIVERSITY OF GHANA



DEPARTMENT OF GEOGRAPHY AND RESOURCE DEVELOPMENT

SPATIAL ACCESSIBILITY TO MAJOR HEALTHCARE FACILITIES IN ADA EAST DISTRICT USING GEOGRAPHIC INFORMATION SYSTEMS TECHNIQUES

BY

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A LONG ESSAY SUBMITTED TO THE DEPARTMENT OF GEOGRAPHY AND RESOURCE DEVELOPMENT, UNIVERSITY OF GHANA, IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF ARTS IN GEOGRAPHY AND RESOURCE DEVELOPMENT

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DECLARATION

I hereby declare that this long essay, undertaken under the supervision of Dr. Ebenezer Nikoi, is the result of my own original research and work. It is submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Arts in Geography and Resource Development at the University of Ghana, Legon. This work has not been submitted for any other degree or examination at any other university, either in whole or in part, except where due acknowledgement has been made in the text.

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DEDICATION

This work is dedicated to my family, whose support made this study possible, and to the households of Ada East District who welcomed a stranger into their homes and shared their time and their struggles so that this research could be undertaken.

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I am deeply grateful to my supervisor, Dr. Ebenezer Nikoi, for the guidance, patience, and constructive criticism that shaped this study from its earliest stages to its completion. I also extend my sincere appreciation to his Teaching Assistant, Endurance Nyame, who carefully reviewed successive drafts of this work and provided valuable suggestions for improvement before I submitted them to my supervisor.

I am especially indebted to Dr. Ebenezer Apaflo for his generous assistance during the fieldwork. He arranged my daily schedules and identified respondents in every community I visited, thereby facilitating the successful conduct of the household survey.

I thank the Ada East District Health Directorate for their cooperation and for the information made available during this research. I am equally grateful to the staff of the Ada East District Hospital, Kasseh Health Centre, and Ada Health Centre for their assistance.

I remain particularly thankful to the seventy households across Ada East District who consented to be interviewed for this study. Their readiness to share their experiences of accessing healthcare, often under challenging circumstances, forms the foundation of this research.

I also acknowledge the friends who assisted with interpretation during the fieldwork in Ada. Their support was essential in ensuring effective communication with respondents.

Finally, I express my heartfelt gratitude to my family and friends for their constant encouragement and support throughout the period of this study.

ABSTRACT

Access to healthcare remains geographically uneven in sub-Saharan Africa, and riverine and island communities face barriers that conventional road-based accessibility models often miss. This study assessed spatial accessibility to major healthcare facilities in Ada East District, Greater Accra Region, Ghana, where the Volta River, its distributaries, and the Songor Lagoon leave many communities reachable only by water. A quantitative, cross-sectional design combined a structured household survey of 70 respondents in seven purposively selected communities (mainland, riverbank, and island or peninsula) with multi-modal GIS network analysis in QGIS that incorporated road and digitised boat-crossing links. Data were analysed using chi-square tests of association and Cramer's V.

Settlement type was strongly associated with self-reported travel time ($V = 0.75$), transport cost ($V = 0.79$), perceived access ($V = 0.68$), and perceived harm to health from delays in reaching care ($V = 0.50$). Island and peninsula households reported the longest journeys, the highest costs, and the greatest dependence on boat crossings. NHIS registration was not significantly associated with forgoing care, and forgoing care showed no settlement-type gradient. Incomplete topological integration of the boat links meant that the GIS model returned no usable travel times for island and peninsula households. The study concludes that the presence of a water crossing is the dominant driver of accessibility difficulty and that insurance coverage alone does not offset it. It recommends targeted water transport interventions, a review of non-financial barriers to care-seeking, and completion of the GIS network model.

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CHAPTER ONE
INTRODUCTION

1.1 Background to the Study

Access to healthcare is widely recognised as a fundamental human right and a key determinant of whether people live healthy, productive lives. The World Health Organization (WHO) is explicit on this point: universal health coverage cannot be achieved unless people can get the care they need without being pushed into financial hardship (WHO, 2018). Nearly five decades after the Alma-Ata Declaration of 1978 committed nations to health for all, that promise still falls short in much of sub-Saharan Africa, particularly for populations living furthest from the health system's reach.

The challenge is especially acute in rural and peri-urban areas, where distance, poor roads, and inadequate facility supply combine to create barriers that prevent people from reaching services in time. Globally, approximately half the world's population lacks access to essential health services, and around 100 million people are pushed into extreme poverty each year by out-of-pocket health costs (WHO & World Bank, 2021). In sub-Saharan Africa, Gething et al. (2012) found, using a national-scale GIS audit in Ghana, that a third of women lived beyond the clinically significant two-hour threshold for reaching emergency obstetric care, with the gap widest in the most remote regions. This pattern is consistent with earlier findings documented across the region (Okafor, 1990).

Ghana has made real strides in improving healthcare access over the past two decades. The National Health Insurance Scheme (NHIS), introduced in 2003, reduced financial barriers to care, and the Community-based Health Planning and Services (CHPS) programme brought basic preventive and curative services closer to village-level populations. Despite these gains, geographic inequalities remain significant. National data show that while approximately 92

percent of urban residents live within one kilometre of some form of health facility, only 35 percent of rural dwellers do (Ghana Statistical Service, 2021). The gap is wider still for major healthcare facilities, that is, hospitals and polyclinics, which are the only tier capable of handling emergencies, complicated deliveries, and surgical cases.

Ada East District in the Greater Accra Region presents a particularly compelling case of these challenges. Situated at the mouth of the Volta River, the district is characterised by a fragmented physical geography in which more than thirty communities sit on islands and peninsulas accessible only by water. For these communities, reaching any mainland health facility requires crossing the river first. This makes straight-line distance a profoundly misleading measure of access: a community two kilometres across a lagoon may face a harder journey than one fifteen kilometres away on a connected road network. GIS-based travel time modelling, which can incorporate both road and water transport, is the appropriate tool for capturing this reality. Yet no such assessment has previously been conducted in Ada East District.

The present study therefore combines household survey data with multi-modal GIS network analysis to assess spatial accessibility to major healthcare facilities in the district. By documenting both self-reported travel experiences and GIS-modelled travel times, the study aims to provide planners with evidence that reflects the true cost of reaching care for island and peninsula communities.

1.2 Problem Statement

The fundamental problem in Ada East District is not the complete absence of health facilities but the unequal spatial distribution of major healthcare facilities relative to where people live. CHPS compounds and health posts exist across the district, but these are designed

for first-contact primary care only. They cannot handle surgical emergencies, complicated deliveries, or patients requiring in-patient care. When a community member needs that level of care, they must reach a major facility, and in Ada East that journey is frequently long, costly, and physically hazardous.

For the purposes of this study, major healthcare facilities are defined as facilities that have emergency care capacity, in-patient services, and a resident qualified clinician, whether a medical doctor or a Physician Assistant, on staff. In Ada East District these comprise the Ada East District Hospital (known locally as Faith Kope), Kasseh Health Centre, and Ada Health Centre (Ada Foah). Lower-tier CHPS compounds and health posts are excluded because they lack the capacity to manage the conditions that constitute the focus of this analysis. Staffing at these facilities, confirmed directly with the Ada East District Health Directorate, follows the expected tier structure of the Ghanaian health system: the Ada East District Hospital is led by a resident medical doctor, while both Kasseh Health Centre and Ada Health Centre (Ada Foah) are Physician Assistant-led, as is typical for the health centre tier nationally. All three nonetheless meet the definition above and were retained as destination facilities on that basis.

For island and peninsula communities, the journey first requires crossing the Volta River or its distributaries. Boat fares, waiting times, and the risk of bad weather or engine failure substantially increase the time cost of these journeys (Thaddeus & Maine, 1994; Dotse-Gborgbortsi et al., 2022). Transport costs represent a further financial burden for predominantly fishing and farming households with low and irregular incomes. Research in Ghana and sub-Saharan Africa also documents that when people have negative experiences with nearby facilities, they bypass them in favour of facilities they trust more, even at greater cost and distance (Kruk et al., 2009).

Existing research establishes several things. Geographic distance to care is consistently associated with lower utilisation, delayed care-seeking, and poorer outcomes, in Ghana as elsewhere (Buor, 2003; Gething et al., 2012; Dotse-Gborgbortsi et al., 2022). Network-based GIS methods have replaced straight-line distance as the standard way of measuring access, although validation studies show that modelled travel times tend to understate the times that patients actually report (Rudolfson et al., 2020). Non-spatial factors, including cost, perceived quality, and insurance, also shape care-seeking independently of distance (Kruk et al., 2009; WHO & World Bank, 2021).

What is not yet known is how these relationships operate where the main barrier is not distance along a road but a water crossing. Accessibility studies in Ghana have largely measured access to any formal facility rather than to major facilities with emergency and in-patient capacity, and they have not addressed water transport. Three questions therefore remain open: how a household's position relative to a water crossing differentiates travel time, cost, and perceived access to major facilities in a district such as Ada East; whether insurance coverage and other non-spatial enabling factors offset that geographic disadvantage or leave it intact; and how far self-reported and GIS-modelled travel times diverge where boat transport is involved. This study addresses these three questions by combining household survey data with multi-modal GIS network analysis that attempts to incorporate both road and boat transport modes. It contributes empirical evidence on how a water crossing structures access to major facilities, and methodological evidence on the limits of road-and-boat network models whose boat links are not fully integrated.

1.3 Research Objectives

The overarching objective of this study is to assess spatial accessibility to major healthcare facilities in Ada East District, Ghana, and to identify the geographic and social factors that explain patterns of access and non-use. The study pursues the following three specific objectives:

1. To map the spatial distribution of major health facilities in Ada East District in relation to the location of surveyed population settlements, establishing the broad geographic pattern of facility access that the household survey and hypothesis testing subsequently quantify by settlement type.
2. To analyse self-reported travel times, transport modes, and transport costs from surveyed settlements to the nearest major health facility, examine how these accessibility measures vary by settlement type, and complement these findings with GIS-modelled travel times derived from a multi-modal (road + boat) network.
3. To investigate non-spatial factors, specifically perceived access to care, transport cost, and insurance coverage, and to examine how these factors interact with settlement type to influence healthcare-seeking behaviour in Ada East District.

1.4 Hypothesis

The hypothesis is formulated to be directly testable using chi-square tests of association on the household survey data, and is examined in relation to the study's core measures of spatial accessibility.

Null Hypothesis (H_0): There is no statistically significant relationship between settlement type (mainland, riverbank, or island/peninsula) and spatial accessibility to major

healthcare facilities, whether measured by self-reported travel time, transport cost, or perceived access, in Ada East District.

Alternative Hypothesis (H_1): There is a statistically significant relationship between settlement type and spatial accessibility to major healthcare facilities, such that island and peninsula communities experience significantly longer travel times, higher transport costs, and poorer perceived access than mainland and riverbank communities.

1.5 Significance of the Study

For the Ada East District Health Directorate, this study may provide actionable spatial data to inform decisions about facility location, mobile health deployment, and outreach priorities. For the Greater Accra Regional Health Directorate, the findings are expected to contribute to an understanding of how geographic barriers in coastal and estuarine districts affect service utilisation and equity, and to demonstrate why road-only accessibility measures are insufficient for planning in water-fragmented districts.

For the academic community, the study makes both an empirical and a methodological contribution. It documents healthcare accessibility in an under-researched district and demonstrates how GIS analysis can be adapted to incorporate water transport modes, an approach relevant to riverine and coastal communities across West Africa. The comparison of self-reported and GIS-modelled travel times further highlights the limitations of pure network models when boat crossings are required. For local communities, the findings may support advocacy by providing empirical documentation of disadvantage that can be used in engagements with district assemblies and health authorities.

1.6 Scope and Limitations

This study is geographically limited to Ada East District, Greater Accra, Ghana. The study focuses on major healthcare facilities defined as those with emergency care capacity, in-patient services, and qualified doctors on staff. In practice, these are the Ada East District Hospital, Kasseh Health Centre, and Ada Health Centre (Ada Foah). CHPS compounds and health posts are excluded. The temporal scope covers 2025 to 2026, using 2021 Population and Housing Census data, with primary fieldwork conducted in May 2026.

The study acknowledges several limitations. First, the household survey produced 70 completed responses across seven purposively selected communities rather than the larger random sample originally envisioned. This resulted from fieldwork logistics in water-fragmented terrain and the absence of ethical clearance for facility-based data collection (Section 3.5). Second, while a multi-modal GIS network (roads + boat crossings) was constructed and used to model travel times, the boat crossing segments were not fully topologically integrated with the road network at every landing point. As a result, the network analysis returned null or zero travel times for the island and peninsula households. These null values are an artefact of incomplete network topology rather than a measurement of travel time, and they are reported here as an unresolved methodological limitation. They are consistent with the water barrier documented by the survey, but they do not by themselves demonstrate it; self-reported times remain the primary and only valid measure of accessibility for these communities. Third, the convenience sample of respondents is not adequate for district-wide statistical inference, and findings should be interpreted as indicative patterns within the sampled communities. Fourth, women accounted for 84.3 percent of respondents, a consequence of deliberately targeting women, who are mostly the household caretakers. Reported care-seeking behaviour, perceived barriers, and unmet need may therefore reflect

women's experience of the health system more closely than that of the household as a whole. Fifth, the questionnaire named only the Ada Health Centre at Ada Foah (listed as "Ada Foah Government Hospital") together with "Other" and "Private clinic", so use of the Ada East District Hospital and Kasseh Health Centre could not be identified and travel time and cost could not be compared facility by facility. Distances from household GPS points to facilities (Section 4.3.4) are straight-line and are based on published facility coordinates.

1.7 Justification of the Study

This study is justified on three grounds. First, it addresses a documented evidence gap: the Ada East District Health Directorate lacks spatially referenced data identifying which communities have adequate access to major health facilities. By producing GIS-based spatial distribution maps, multi-modal travel time surfaces, and documenting self-reported travel patterns, the study provides a planning baseline.

Second, the study is relevant to national policy priorities. Ghana's health sector strategic plans consistently emphasise reducing geographic inequities in access to services. At the district level, implementing these priorities requires localised spatial analysis that national frameworks cannot supply. Third, the study contributes methodologically to the literature on healthcare accessibility in water-fragmented environments, where standard road-only GIS models systematically underestimate access difficulties.

CHAPTER TWO
LITERATURE REVIEW AND THEORETICAL FRAMEWORK

2.1 Theoretical Framework

Two theoretical frameworks guide this study. Andersen's Behavioural Model of Healthcare Utilisation explains how predisposing, enabling, and need factors combine to shape care-seeking, and the Three Delays Model identifies where, between deciding to seek care and receiving it, spatial barriers do their damage. The review of access concepts, empirical studies, the Ghanaian health system, and approaches to measuring spatial accessibility that follows is read in light of these frameworks.

2.1.1 Andersen's Behavioural Model of Healthcare Utilisation

This study is guided by Andersen's Behavioural Model of Healthcare Utilisation (Andersen, 1968, 1995). The model proposes that healthcare utilisation is shaped by three sets of factors: predisposing factors (demographic and social characteristics that influence the propensity to seek care), enabling factors (resources that make care possible, including spatial accessibility, transport, and insurance), and need factors (perceived and evaluated health status). Spatial accessibility is treated as a community-level enabling factor. The model is particularly useful for the present study because it draws attention to the interaction between spatial and non-spatial enabling factors, for example how the combination of long travel times and high transport costs may outweigh the enabling effect of NHIS coverage.

2.1.2 The Three Delays Model

The second framework is the Three Delays Model developed by Thaddeus and Maine (1994). The model identifies three sequential delays that contribute to maternal mortality and, by extension, to poor outcomes for other time-sensitive conditions: the delay in deciding to

seek care, the delay in reaching an appropriate facility, and the delay in receiving adequate care once at the facility. The second delay is most directly linked to spatial accessibility and is the primary analytical focus of this study. In Ada East District, this delay has a compound structure: for island and peninsula communities it involves both a water segment and a subsequent road segment, each of which introduces its own time, cost, and risk.

2.1.3 Conceptual Framework

Figure 1 presents the conceptual framework that guides the empirical analysis. Spatial accessibility, determined by settlement location relative to major facilities, the structure of the transport network (including boat crossings), and the resulting travel time and cost, interacts with non-spatial enabling factors (perceived quality, insurance coverage, household resources) and predisposing characteristics to produce healthcare-seeking behaviour and, ultimately, health outcomes. The framework also acknowledges the feedback loop whereby negative experiences of delay and cost may reinforce reliance on traditional healers and faith healing, further reducing formal facility utilisation.

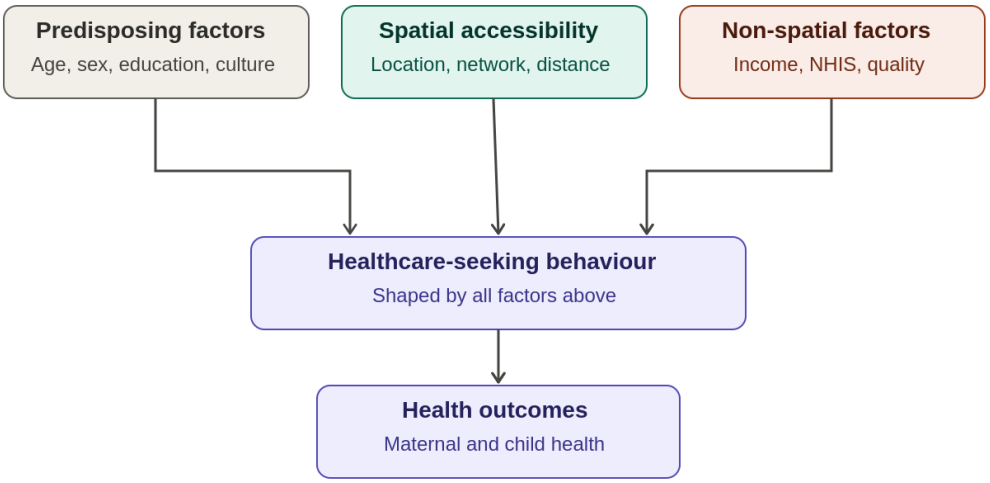


Figure 1: *Conceptual Framework: Interaction Between Spatial Accessibility, Non-Spatial Enabling Factors and Healthcare-Seeking Behaviour in Ada East District*

2.2 The Concept of Access to Healthcare

Access to healthcare is one of the most frequently invoked but inconsistently defined concepts in health geography and health systems research. The most influential framework remains that of Penchansky and Thomas (1981), who proposed five interrelated dimensions: availability, accessibility, accommodation, affordability, and acceptability. Availability refers to whether enough facilities and providers exist to meet demand. Accessibility in the spatial sense refers to whether people can physically reach services given their location and available transport options. Accommodation describes whether services are organised to fit users' practical circumstances, including opening hours and appointment systems. Affordability measures whether care costs are manageable relative to household resources. Acceptability captures whether people trust and feel comfortable with the providers and the style of care offered.

This framework remains widely cited but has been criticised for being largely supply-side in orientation. Levesque et al. (2013) argued that a genuinely complete model of access must also incorporate demand-side factors such as health literacy, the ability to perceive need, and the capacity to seek care. They proposed a patient-centred framework distinguishing five supply-side dimensions and five corresponding demand-side abilities. This critique is important for the current study because it makes visible how structural disadvantage compounds across all dimensions simultaneously in communities like those in Ada East, where geographic isolation, low incomes, limited insurance coverage, and low formal education interact. The current study draws on Penchansky and Thomas to operationalise survey variables and on Levesque et al. to interpret findings, particularly where spatial and non-spatial barriers compound each other.

More recent frameworks incorporate a temporal dimension, distinguishing between potential access (the structural conditions that make care possible) and realised access (actual service use). GIS analysis primarily measures potential access, while household surveys can partially illuminate the gap between potential and realised access. This study employs both approaches to capture the structural geography of access alongside the lived experience of reaching care.

2.3 Empirical Studies on Healthcare Access

This section reviews empirical studies of healthcare access in three parts: the evidence on distance and health outcomes, the evidence on non-spatial determinants of care-seeking, and what is known about access in Ada East District itself.

2.3.1 Distance and Health Outcomes

The relationship between distance to health facilities and health outcomes is one of the most consistently documented findings in health geography. Greater distance is associated with lower facility utilisation, delayed care-seeking, and worse health outcomes across maternal health, child health, chronic disease management, and emergency care (Gething et al., 2012; Dotse-Gborgbortsi et al., 2022). The conceptual foundation for much of this literature was established by Thaddeus and Maine (1994) in their Three Delays Model, which identified the second delay, reaching an appropriate facility, as a critical contributor to maternal mortality.

Evidence from elsewhere in sub-Saharan Africa points the same way. Okafor (1990) examined the spatial dimensions of accessibility to general hospitals in rural Nigeria. Kruk et al. (2009) found that women in rural Tanzania bypassed nearer primary care facilities for childbirth in favour of facilities they trusted more, and validation work in Rwanda found that patient-reported travel times to a district hospital exceeded modelled times by a factor of about 1.5 (Rudolfson et al., 2020). Taken together, these studies indicate that distance matters, but that the burden of reaching care is only partly captured by modelled distance.

In Ghana specifically, Buor (2003) demonstrated that distance was among the strongest predictors of health facility utilisation even after controlling for socioeconomic variables. More recent Ghanaian studies extend these findings. Boakye et al. (2026) used Euclidean buffer analysis in the Twifo Ati-Morkwaa District and found that about 95 percent of sampled communities lay within 5 km of a primary healthcare facility in straight-line terms, yet transport costs, poor road conditions, and physical barriers substantially constrained effective access, which shows how straight-line measures can overstate access, while Dotse-Gborgbortsi et al. (2022) found that self-reported distance to maternal health services in Ghana was often considerably greater than GIS-modelled distance, particularly for the most remote households.

International thresholds provide useful benchmarks: the WHO recommends that primary care should be reachable within 30 minutes (WHO, 2010), and the widely applied standard for emergency obstetric care is two hours. These thresholds inform the accessibility classification framework used in the present study.

2.3.2 Non-Spatial Determinants of Healthcare Utilisation

Spatial accessibility is necessary but not sufficient for healthcare utilisation. Economic barriers include both the direct costs of care and the indirect costs of transport and lost income. Even where the NHIS nominally covers services, out-of-pocket costs remain a major driver of financial hardship for low-income households (WHO & World Bank, 2021). Quality perceptions also matter: research across sub-Saharan Africa shows that people frequently bypass the nearest facility when they perceive it as poorly staffed, poorly equipped, or disrespectful, preferring to travel further to facilities they trust (Kruk et al., 2009). Cultural and social factors, including gender norms, religious beliefs, and the availability of traditional healers, further influence care-seeking independently of distance.

Taken together, the studies reviewed in this section tend to treat distance and non-spatial determinants as separate influences on care-seeking. Studies that combine household evidence on cost, insurance, and perceived access with a spatial measure of travel burden, in a setting where water transport is involved, were not found in the literature reviewed for this study.

2.3.3 Healthcare Access in Ada East District

Ada East District combines a distinctive riverine geography with a dispersed population of approximately 75,000 people spread over roughly 1,100 square kilometres, and an economy built largely on artisanal fishing, salt harvesting, and subsistence farming. Health indicators in

the district lag behind Greater Accra's regional averages, and NHIS coverage falls below the national average, partly because registering or renewing membership requires travel to district offices that are difficult for island and peninsula residents to reach. Before the present study, no systematic GIS-based assessment of accessibility to major facilities had been conducted in the district.

2.4 Ghana's Health System and the Role of Major Facilities

Ghana's health system is organised in a tiered hierarchy, from CHPS compounds at the base through health centres and polyclinics to district, regional, and teaching hospitals at the apex. CHPS compounds are designed to deliver basic preventive and curative services at the community level. Health centres and hospitals provide progressively higher levels of care, including emergency services, in-patient admission, and surgical capacity.

For this study, major healthcare facilities are defined as those at the health-centre level and above that have emergency care capacity, in-patient beds, and medical doctors on staff. In Ada East District the relevant facilities meeting this definition are the Ada East District Hospital, Kasseh Health Centre, and Ada Health Centre (Ada Foah). The spatial distribution of major facilities across Ghana is heavily skewed toward urban centres and district capitals, leaving many rural and peri-urban populations with long journeys to reach this tier of care.

2.5 Measurement of Spatial Accessibility

Spatial accessibility has been operationalised through a range of methodological approaches, increasing in sophistication. The simplest is proximity measurement, calculating straight-line (Euclidean) distance to the nearest facility. Proximity measures are computationally easy and require minimal data, but they are widely acknowledged to be poor

proxies for actual accessibility because they ignore the structure of the transport network, physical barriers such as rivers, and variation in travel speeds (Neutens, 2015; Higgs et al., 2019).

Network analysis, which calculates travel time or distance along actual road, path, or waterway networks while accounting for variation in travel speed by road class, is more realistic and has become the standard approach in GIS-based healthcare accessibility research. The Two-Step Floating Catchment Area (2SFCA) method developed by Luo and Wang (2003) further refines this by incorporating the supply-to-demand ratio at each facility, capturing not only distance but also how crowded a facility is relative to the population it serves.

For Ada East District, the most significant methodological challenge is the role of water barriers. Standard GIS road network analyses assume that movement is constrained to road networks and cannot account for water transport without deliberate modification. Validation studies comparing GIS-modelled travel times against patient-reported travel times in other low- and middle-income settings have found that standard road-network models systematically underestimate true travel time; in the largest such validation study to date, patient-reported travel times to a district hospital in Rwanda exceeded AccessMod-modelled times by a factor of approximately 1.5 (Rudolfson et al., 2020). This underestimation is expected to be more pronounced where a water crossing is required, since standard road-network tools do not model boat transport without deliberate modification (Ray & Ebener, 2008). This study therefore constructs a multimodal network that includes both roads and digitised boat-crossing links. At the same time, it acknowledges that incomplete topological integration of the boat segments limits the model's ability to return continuous travel times for every island household, a limitation that itself becomes evidence of the structural barrier under investigation.

2.6 GIS Applications in Healthcare Accessibility Analysis

Geographic Information Systems have become a standard tool in health geography for facility planning, equity analysis, and accessibility assessment. GIS enables researchers to combine facility location data, population distribution, and transport networks in order to produce maps and quantitative measures of accessibility that would be difficult or impossible to generate by other means.

GIS-based studies of healthcare accessibility in Ghana have been conducted in several districts, including Atwima Mponua and parts of the Northern Region. Most of these studies, however, focus on access to any formal facility rather than specifically to major facilities with emergency and in-patient capacity. None have fully grappled with the water transport challenge that characterises Ada East. International studies in Bangladesh, the Philippines, and parts of Indonesia have adapted GIS methods to model water-based transport, consistently finding that water barriers produce access disadvantages that cannot be predicted from road-based distance alone. The present study builds on this body of work by constructing a multi-modal network for Ada East and by comparing the resulting modelled times with self-reported experiences.

2.7 Research Gap and Contribution of the Study

The literature reviewed above establishes that distance matters, that network-based GIS measures are preferable to proximity measures, and that cost, quality perceptions, and insurance also shape care-seeking. Three gaps remain. First, accessibility studies in Ghana have mostly assessed access to any formal facility rather than to major facilities with emergency and in-patient capacity, and they have not addressed water transport (Section 2.6). Second, work on non-spatial determinants seldom examines how those factors interact with

spatial position, so it is unclear whether insurance coverage offsets or leaves intact geographic disadvantage in water-fragmented settings (Section 2.3.2). Third, the validation studies reviewed here, which compare modelled and self-reported travel times (Dotse-Gborgbortsi et al., 2022; Rudolfson et al., 2020), did not focus on settings where a water crossing is required.

By linking household survey evidence to a multi-modal network model in Ada East District, this study addresses these gaps. Drawing on Andersen's model and the Three Delays Model, it tests the expectation that a water crossing is a distinct source of the second delay and that insurance coverage does not remove it.

CHAPTER THREE
STUDY AREA AND METHODOLOGY

3.1 Study Area

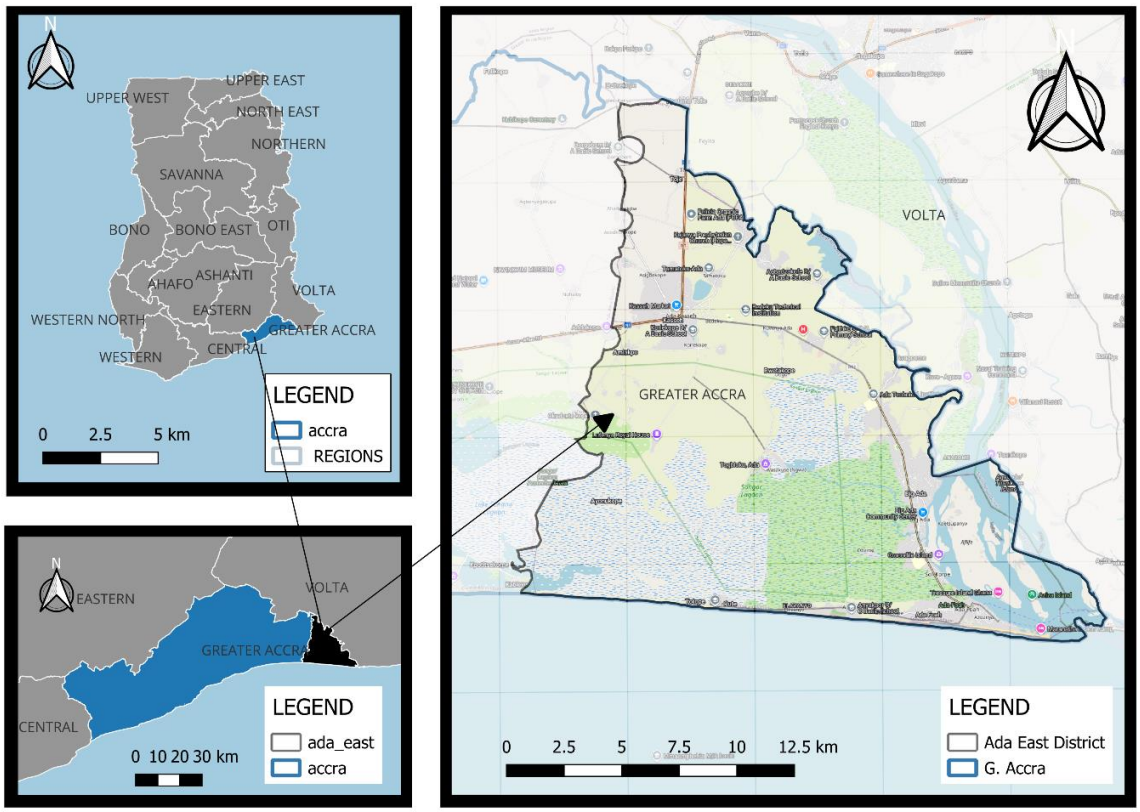


Figure 2: *Study Area Map of Ada East District, Greater Accra Region, Ghana*

3.1.1 Location and Administrative Context

Ada East District is one of the sixteen districts of the Greater Accra Region of Ghana. It is located at the mouth of the Volta River and is bounded to the north by Shai Osudoku District, to the west by Ningo Prampram District, and to the east and south by the Atlantic Ocean. The district capital is Ada Foah. The district covers approximately 1,100 square kilometres and has a population of approximately 75,000 according to the 2021 Population and Housing Census (Ghana Statistical Service, 2021).

3.1.2 Physical Geography and Implications for Healthcare Access

The physical geography of Ada East is defined by the lower Volta estuary. Multiple distributaries, backchannels, and the Songor Lagoon divide the district into geographic fragments. More than thirty communities have no direct road connection to the district capital and depend on boats as their primary means of transport. Additional communities have road connections that become impassable or severely degraded during the rainy season. This physical fragmentation is the central geographic fact that shapes healthcare accessibility in the district and motivates the multi-modal modelling approach adopted in this study.

3.1.3 Population and Socioeconomic Context

The population of Ada East is predominantly engaged in fishing, salt harvesting, and small-scale agriculture. Educational attainment is lower than the national average, and NHIS coverage is also below the national average. Household incomes are generally low and irregular, which amplifies the impact of transport costs on the decision to seek care.

3.2 Research Design

This study adopted a quantitative, cross-sectional research design. Primary data were collected through a structured household survey whose items were largely closed-ended and analysed statistically, and spatial data were analysed with GIS network methods to model travel time. Both components are quantitative: GIS spatial analysis (constructing a multi-modal road and boat network and calculating travel times) and survey analysis (descriptive statistics and chi-square tests of association). No qualitative methods of data collection or analysis were used. One open-ended survey item invited respondents to comment on their priorities and challenges; these comments are reported in Section 4.6 as illustrative quotations and were not coded or analysed thematically. Both components were carried out during the May 2026

fieldwork period. Key informant interviews with health workers and community leaders, and exit interviews with patients, were originally planned but did not proceed (Section 3.5) and form no part of the design.

3.3 Data Collection Methods

3.3.1 GIS Spatial Analysis and Multi-Modal Travel Time Modelling

Health facility location data were obtained from the Ghana Health Service District Health Information Management System (DHIMS2), cross-checked against the Ada East District Health Directorate facility register, and verified through GPS surveys conducted during fieldwork. The major facilities used as destinations in the network analysis are the Ada East District Hospital, Kasseh Health Centre, and Ada Health Centre (Ada Foah).

Road network data were obtained from OpenStreetMap and imported into QGIS using the QuickOSM plugin. The road data were cleaned using a sequence of Processing Toolbox tools: Fix geometries, Snap geometries to layer, Multipart to singleparts, and Dissolve. Boat crossing links were manually digitised between known island and peninsula landing sites and corresponding mainland landings. These links were assigned an approximate speed of 10 km/h and a waiting-time penalty, then merged with the cleaned road network to create a multi-modal network layer.

Travel times were calculated using two complementary approaches. First, Service Area (from layer) analysis was run from the three major facilities for thresholds of 30, 60 and 120 minutes, producing the isochrone surfaces presented in Figure 4. Second, an origin-destination matrix was computed using the QNEAT3 plugin, with the 70 household GPS points as origins and the three major facilities as destinations. The minimum total cost for each origin was retained as the modelled travel time to the nearest facility. Where the network returned null or

zero values, predominantly for island and peninsula points, this indicates that the algorithm could not trace a continuous path within the current network topology. Because the digitised boat links were not fully snapped to the road network at every landing point, these returns are treated as a known failure of the model rather than as a travel time estimate, and they are excluded from the quantitative comparison in Section 4.5.

QGIS was selected as the primary GIS platform because it is free, open-source, and supports the required network analysis plugins, making the methodology reproducible in resource-limited settings.

3.3.2 Household Survey

A structured household survey was conducted across seven communities between 14 and 20 May 2026. Data were collected using the KoboCollect mobile application and covered demographic information, self-reported travel time and cost to the nearest major facility, primary transport mode, boat crossing requirement, perceived barriers, NHIS registration status, and experiences of unmet need. The questionnaire is reproduced in Appendix B.

Study population and eligibility. The study population was adults living in households in the sampled Ada East District communities. There was no formal eligibility rule beyond being an adult (aged 18 or over) who was available at home and willing to be interviewed. The researcher deliberately sought to speak with women, because women are mostly the caretakers in the household, and one adult respondent answered on behalf of the household. This is why several items ask about the experience of any household member.

Sampling approach. Seven communities were selected purposively so that all three settlement types would be represented: Kasseh, Totimekope, Ocanseykope, Pute, Azizanya, Aflive, and Aziza. Within each community, households were selected because a respondent

was available at home when the team visited, and the researcher interviewed adults who were available, with a preference for women. A community member accompanied the researcher and helped to collect the data, which encouraged cooperation, and every household approached agreed to be interviewed, so there were no refusals. No household listing was available from which a random sample could be drawn. Settlement type was recorded by the enumerator for each household. Mainland households (19) were in Kasseh (10), Totimekope (6), Ocanseykope (2), and Pute (1); riverbank households (28) were in Azizanya (11), Pute (11), Totimekope (2), and Ocanseykope (4); and island or peninsula households (23) were in Aflive (13) and Aziza (10). Community sample sizes were 13 (Aflive), 12 (Pute), 11 (Azizanya), 10 (Kasseh), 10 (Aziza), 8 (Totimekope), and 6 (Ocanseykope). Interviews were conducted face to face in respondents' homes, with interpretation into local languages where needed. The final sample comprised 70 households. This size reflects what fieldwork logistics in water-fragmented terrain allowed rather than a calculated sample size.

Representativeness. The sample is not statistically representative of the district's population of approximately 75,000, nor of the more than thirty communities that depend on water transport. Allocation across settlement types was not proportional to population, women made up 84.3 percent of respondents because the survey deliberately targeted women as the main household caretakers, and the sample was concentrated in seven communities that could be reached within a one-week fieldwork period. The sample is therefore suited to comparing settlement types within the sampled communities, which is what the hypothesis requires, and not to estimating district-wide prevalence. The p-values reported in Chapter Four should be read in that light (Section 4.4).

3.3.3 Travel Time Classification

Both self-reported and GIS-modelled travel times were classified into four categories following WHO guidelines and established West African practice: good access (less than 30 minutes), moderate access (30–60 minutes), limited access (60–120 minutes), and critical access (more than 120 minutes). These categories are used both for descriptive mapping and for the comparison between modelled and reported times. No household in the sample reported or was modelled at more than 120 minutes, so the critical access category is retained for comparability with the wider literature but is not populated in the results that follow.

3.4 Data Analysis

Spatial data were managed and analysed in QGIS 3.x. Survey data were cleaned and analysed in Python using the SciPy library, with descriptive statistics, chi-square tests of association, and Cramer's V as an effect size measure. Statistical significance was set at $p < .05$. Where expected cell counts were low, categories were collapsed to satisfy the assumptions of the chi-square test. Straight-line distances from each household's GPS point to the Ada Health Centre, the Ada East District Hospital, and Kasseh Health Centre were calculated with the haversine formula, using approximate published facility coordinates (Section 4.3.4).

Outcome variables. Eight survey outcomes were examined, all categorical. The accessibility outcomes tied directly to the hypothesis were self-reported one-way travel time to the nearest major facility (under 30 minutes, 30 to 60 minutes, or one to two hours), one-way transport cost (GHS 60 or less, or more than GHS 60), whether a boat crossing was required (yes or no), and perceived access to healthcare in the community (good, poor, or very poor). The care-seeking outcomes relating to the third objective were whether the household had ever been unable to reach a facility because of transport problems (yes more than once,

yes once, or never), whether a delay in reaching care had ever harmed a household member's health (yes significantly, somewhat, no, or don't know), whether the household currently had an unmet health need because of access barriers (yes or no), and whether a household member had ever chosen not to seek needed care (yes or no). GIS-modelled travel time, classified using the categories in Section 3.3.3, was compared descriptively with self-reported travel time.

Independent variables. The main independent variable was settlement type (mainland, riverbank, or island or peninsula), recorded by the enumerator for each household according to its location relative to the road network and to any water crossing needed to reach a major facility. Two further independent variables were examined: the respondent's education level (no formal education, primary, junior high school, or vocational), as a predisposing characteristic, and NHIS registration status (active, lapsed, or never registered), as an enabling factor, following Andersen's model (Section 2.1.1).

Choice of test and associations examined. Chi-square tests of association were used because both the outcome and independent variables are nominal or ordinal categories, so the test of independence between two categorical variables is appropriate and requires no assumption of normality. Because chi-square p-values depend on sample size and the sample is small and was not randomly selected, Cramer's V was reported alongside each test to show the strength of association on a scale from 0 to 1. Ten associations were tested: settlement type against each of the eight outcomes above; education level against perceived access; and NHIS status against whether a household had forgone needed care. The tests for travel time, transport cost, boat crossing, and perceived access address the hypothesis in Section 1.4; the tests for the remaining outcomes, and those involving education and NHIS status, address the third objective on non-spatial factors. All ten results are summarised in Table 12.

3.5 Ethical Considerations

Ethical approval for the household survey component of the study was granted by the University of Ghana Geography and Resource Development Department. Participation was voluntary, and all respondents provided informed consent before the interview began. GPS coordinates of households were recorded only with explicit consent and were stored separately from personal identifiers.

The original research design also included key informant interviews with health workers and community leaders, exit interviews with patients, and the collection of information from within health facilities. Because these activities involve health facility staff and premises, they require separate ethical clearance from the Ghana Health Service. An application was made to the Ghana Health Service, but no response had been received by the time of writing. As a result, no data were collected from health workers or from within any health facility, the key informant and patient interviews are not reported in this essay, and facility information (location, tier, and staffing) was drawn from DHIMS2 and the Ada East District Health Directorate facility register (Section 3.3.1). The household survey, which took place in community settings, was not affected. The consequences for the analysis are set out in Section 1.6, and these interviews are proposed as future research in Section 5.5.

CHAPTER FOUR: STUDY FINDINGS

4.0 Overview of Analysis

This chapter presents the study findings. Sections 4.1 to 4.4 report the results of the household survey conducted between 14 and 20 May 2026, which produced 70 completed interviews across seven communities. Section 4.5 presents the results of the GIS multi-modal travel time analysis, including the three-panel service area map and the comparison of modelled and self-reported travel times. Section 4.6 reports respondents' own views, drawn from one open-ended survey item. Section 4.7 discusses the combined findings in light of the literature and the theoretical frameworks reviewed in Chapter Two.

4.1 Spatial Distribution of Major Health Facilities and Surveyed Settlements

Figure 3 shows the spatial distribution of the three major healthcare facilities and the GPS locations of the surveyed households. The Ada East District Hospital and Kasseh Health Centre are located in the northern part of the district on the main road network, while Ada Health Centre lies near the coast at Ada Foah. The surveyed settlements form two broad clusters: a northern mainland cluster centred on Kasseh and a southern cluster of island, peninsula, and riverbank communities along the Volta estuary and the Songor Lagoon. This distribution creates a clear structural north–south accessibility gradient. The map plots all surveyed households as a single layer rather than colour-coding them by settlement type; the settlement-type comparisons that this gradient implies are quantified directly in the cross-tabulations and chi-square tests presented in Sections 4.3 and 4.4.

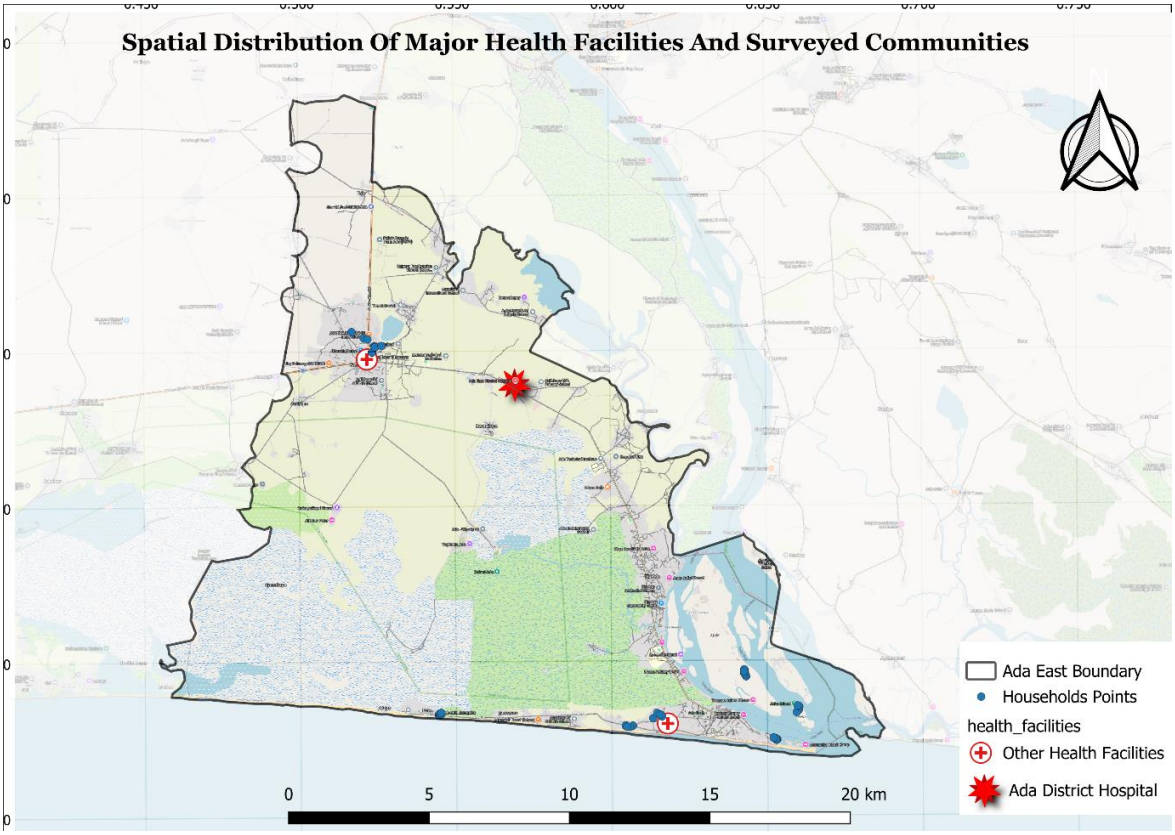


Figure 3: *Spatial Distribution of Major Health Facilities and Surveyed Settlements, Ada East District*

4.2 Demographics of Respondents

Women accounted for 84.3 percent of the 70 respondents. The modal age group was 36–45 years (40 percent of the sample). Sixty percent of respondents reported having no formal education. Fishing was the most common occupation (41.4 percent), followed by farming (17.1 percent). Household sizes were generally large: 65.7 percent of respondents reported living in households of six to nine members. Active membership of the National Health Insurance Scheme stood at 55.7 percent. The settlement-type distribution of the sample was 27.1 percent mainland, 40.0 percent riverbank, and 32.9 percent island or peninsula.

Table 1: Socio-Demographic Characteristics of Respondents (N = 70)

Variable	Category	Frequency	Percent (%)
Sex	Female	59	84.3
	Male	11	15.7
Age group	18–25	1	1.4
	26–35	9	12.9
	36–45	28	40.0
	46–55	16	22.9
	56–65	10	14.3
	Over 65	6	8.6
Education	No formal education	42	60.0
	Primary	24	34.3
	JHS	3	4.3
	Vocational	1	1.4
Occupation	Fishing	29	41.4
	Farming	12	17.1
	Trading	9	12.9
	Unemployed/Homemaker	8	11.4
	Self-employed/Artisan	6	8.6
	Salt harvesting	4	5.7
	Government	1	1.4
	Student	1	1.4
Household size	3–5 people	5	7.1
	6–9 people	46	65.7
	10 or more	19	27.1
NHIS status	Active membership	39	55.7
	Lapsed membership	27	38.6
	Never registered	4	5.7
Settlement type	Mainland	19	27.1
	Riverbank	28	40.0
	Island / Peninsula	23	32.9
Major facility visit, past 12 months	Visited	34	48.6

	No visit	36	51.4
Facility last visited (n = 34)	Ada Health Centre (Ada Foah)	27	79.4
	Other facility	6	17.6
	Private clinic	1	2.9

Source: Field Survey, May 2026. Percentages for facility last visited are based on the 34 households that reported a visit.

4.3 Self-Reported Travel Time, Transport Cost, and Boat Dependency

4.3.1 Self-Reported Travel Time by Settlement Type

Table 2 presents the cross-tabulation of settlement type and self-reported travel time to the nearest major health facility. The pattern is striking. All 23 island and peninsula respondents reported either 30–60 minutes (17.4 percent) or 1–2 hours (82.6 percent); none reported a travel time of under 30 minutes. Mainland respondents showed the opposite pattern: 94.7 percent reported under 30 minutes. Riverbank communities occupied an intermediate position. The association between settlement type and self-reported travel time is highly statistically significant and very strong ($\chi^2 = 79.32$, df = 4, $p < .001$, Cramer's V = 0.75).

Table 2: Settlement Type by Self-Reported Travel Time to Nearest Major Health Facility

Settlement Type	< 30 min	30–60 min	1–2 hours	Total
Mainland	18 (94.7%)	1 (5.3%)	0 (0.0%)	19 (100%)
Riverbank	9 (32.1%)	19 (67.9%)	0 (0.0%)	28 (100%)
Island / Peninsula	0 (0.0%)	4 (17.4%)	19 (82.6%)	23 (100%)

$\chi^2 = 79.32$, df = 4, $p < .001$, Cramer's V = 0.75, N = 70. Source: Field Survey, May 2026.

Tables 2 to 4 report travel time, cost, and boat dependency for the nearest major facility, with the three facilities pooled. Pooling shows how settlement types differ overall, but it cannot show which facility is more accessible to which settlements. Section 4.3.4 takes up that question.

4.3.2 One-Way Transport Cost

Transport cost shows an equally strong relationship with settlement type. All 23 island and peninsula respondents (100 percent) reported paying more than GHS 60 for a one-way journey to the nearest major facility. In contrast, only 5.3 percent of mainland respondents and 28.6 percent of riverbank respondents reported costs above this threshold. The association is highly significant ($\chi^2 = 43.16$, df = 2, $p < .001$, Cramer's V = 0.79).

Table 3: Settlement Type by One-Way Transport Cost (Collapsed to Two Categories)

Settlement Type	GHS 60 or less	More than GHS 60	Total
Mainland	18 (94.7%)	1 (5.3%)	19 (100%)
Riverbank	20 (71.4%)	8 (28.6%)	28 (100%)
Island / Peninsula	0 (0.0%)	23 (100%)	23 (100%)

$\chi^2 = 43.16$, df = 2, $p < .001$, Cramer's V = 0.79, N = 70. Source: Field Survey, May 2026.

4.3.3 Boat Crossing Dependency

The requirement for a boat crossing is almost perfectly predicted by settlement type ($\chi^2 = 61.67$, df = 2, $p < .001$, Cramer's V = 0.94). Every island and peninsula respondent reported that a boat crossing was necessary to reach a major facility. Only one mainland respondent (5.3 percent) and one riverbank respondent (3.6 percent) reported the same. It should be noted that boat dependency is close to definitional for island and peninsula settlements, so the very large effect size here (Cramer's V = 0.94) is better read as confirmation that the settlement-type classification captured the intended geographic distinction than as a substantive finding in its own right.

Table 4: Settlement Type by Boat Crossing Requirement

Settlement Type	Yes	No	Total
Mainland	1 (5.3%)	18 (94.7%)	19 (100%)
Riverbank	1 (3.6%)	27 (96.4%)	28 (100%)
Island / Peninsula	23 (100%)	0 (0.0%)	23 (100%)

$\chi^2 = 61.67$, $df = 2$, $p < .001$, Cramer's $V = 0.94$, $N = 70$. Source: Field Survey, May 2026.

4.3.4 Differentiated Access by Facility

The three major facilities do not offer equivalent access. The Ada East District Hospital and Kasseh Health Centre lie in the northern part of the district on the main road network, whereas Ada Health Centre is on the coast at Ada Foah, where the boat crossing to the island and peninsula communities begins (Plate 3). The Ada East District Hospital is led by a resident medical doctor, while Kasseh Health Centre and Ada Health Centre are Physician Assistant-led (Section 1.2). From a household's point of view the facilities therefore differ in three ways: their position on the road network, their position relative to a water crossing, and their clinical tier and staffing. A household's access to a given facility depends on where it sits relative to all three. In the modelled service areas (Figure 4), the reach of the facilities does not extend across the water to the island and peninsula communities at any of the thresholds used.

The survey recorded which facility a household had used most recently, for households that reported a visit to a major facility in the previous 12 months. Thirty-four of the 70 households (48.6 percent) reported such a visit, and the share differed by settlement type: 7 of 23 island and peninsula households (30.4 percent), 9 of 19 mainland households (47.4 percent), and 18 of 28 riverbank households (64.3 percent). This difference falls just short of significance ($\chi^2 = 5.81$, $df = 2$, $p = .055$, Cramer's $V = 0.29$). Island and peninsula households were thus the least likely to have used a major facility at all, even though every one of them reported unmet health need (Table 8). Among the 34 households that had visited, 27 (79.4

percent) named the Ada Health Centre at Ada Foah, which the questionnaire listed as “Ada Foah Government Hospital”. These included all 7 island and peninsula households, 14 of 18 riverbank households (77.8 percent), and 6 of 9 mainland households (66.7 percent). Six households named another facility (four riverbank and two mainland) and one a private clinic. The reasons given for choosing the facility (multiple responses) were that it was the nearest major facility (25 households, 73.5 percent), referral (17, 50.0 percent), better doctors or equipment (10, 29.4 percent), recommendation by others (6), NHIS coverage being accepted there (5), and being the only facility able to treat the condition (3).

The survey data show a clear pattern in where households go. The facility used by most households, and by every island and peninsula household that had visited, was the Ada Health Centre at Ada Foah. That is the facility at the coastal point where the boat crossing from the island and peninsula communities begins (Plate 3). All seven island and peninsula households that had visited chose this option, so none used the “Other” category. The questionnaire did not list the Ada East District Hospital (Faith Kope) or Kasseh Health Centre separately: its options were “Ada Foah Government Hospital”, “Other”, and “Private clinic”, so any use of those two facilities falls under “Other” (six households, four riverbank and two mainland) and cannot be identified. Travel time and cost could therefore not be compared facility by facility from the survey itself. The household GPS coordinates allow a partial answer, described next.

Distance from households to each facility. GPS coordinates were recorded for all 70 households, although the point for one Aflive household was missing from the exported data and was entered manually from the field record (5°48′0.77″N, 0°38′26.47″E). Straight-line distances from each household to the three major facilities were calculated using the haversine formula. For island and peninsula households (n = 23), the median distances were 1.8 km to

the Ada Health Centre (range 1.7 to 3.0 km), 13.3 km to the Ada East District Hospital (12.6 to 15.4 km), and 17.6 km to Kasseh Health Centre (17.0 to 19.8 km). For riverbank households (n = 28), the medians were 6.5 km (2.3 to 9.8 km), 12.5 km (12.1 to 15.8 km), and 14.5 km (12.9 to 20.0 km) respectively. For mainland households (n = 19), the medians were 17.7 km (1.9 to 18.8 km) to the Ada Health Centre, 6.1 km (5.0 to 13.0 km) to the District Hospital, and 1.1 km (0.3 to 16.5 km) to Kasseh Health Centre. The Ada Health Centre was the nearest facility for all 23 island and peninsula households, all 28 riverbank households, and 9 of the 19 mainland households (those in Totimekope, Ocanseykope, and Pute). Kasseh Health Centre was nearest for the 10 mainland households in Kasseh, whose median distance to it was 0.5 km compared with 5.2 km to the District Hospital. The District Hospital was the nearest facility for no household.

What accounts for the differences. Three points follow. First, the settlement-facility pattern is sharply divided. Island, peninsula, and riverbank households (51 of 70) are all nearest to the Ada Health Centre at Ada Foah and lie between 12 and 20 km from both the District Hospital and Kasseh Health Centre, while the Kasseh households are close to Kasseh Health Centre and 5 km from the District Hospital. Second, the doctor-led District Hospital is the nearest major facility for no one. For island and peninsula households any journey to it would also require the boat crossing, so these households rely in practice on the Physician Assistant-led health centre at Ada Foah, which is consistent with the survey finding that every island and peninsula household that had visited a facility used it. Third, straight-line distance does not explain the burden of travel. Island and peninsula households are the closest group to a major facility, a median of 1.8 km from the Ada Health Centre, yet they reported the longest journeys (19 of 23 reported one to two hours), the highest costs, and universal dependence on boat

crossings (Tables 2 to 4). Differences between settlement types in access to major facilities are therefore explained by the water crossing, together with where each facility sits relative to the road network and the communities it serves, and not by distance alone. These figures are straight-line distances and ignore roads and boat routes, so they indicate relative position, not travel time. Facility locations are based on published coordinates.

4.4 Non-Spatial Factors, Hypothesis Testing, and Coping Strategies

Beyond the core travel time, cost, and boat-crossing measures reported in Section 4.3, the survey also asked respondents about the severity of the barriers they face and how those barriers affect their households. This section reports cross-tabulations of settlement type against five further outcomes: whether a household had ever been unable to reach a facility because of transport problems, how respondents rated their overall access to care, whether a delay in reaching care had ever harmed a health outcome, whether the household currently had an unmet health need, and whether the household had ever chosen to forgo needed care altogether. Chi-square tests of association are reported for each, together with Cramer's V as a measure of effect size. Because respondents were selected by convenience rather than randomly, these tests should be read as describing the strength of association within the seven sampled communities; the resulting p-values do not support inference to the district population as a whole. Where more than 20 percent of cells in a contingency table have an expected count below 5, a standard rule of thumb for chi-square reliability, this is noted and the result is treated as suggestive rather than conclusive.

Table 5: Settlement Type by Whether Household Was Ever Unable to Reach a Facility Due to Transport Problems

Settlement Type	Yes, more than once	Yes, once	Never	Total
Mainland	2 (10.5%)	4 (21.1%)	13 (68.4%)	19 (100%)
Riverbank	15 (53.6%)	6 (21.4%)	7 (25.0%)	28 (100%)
Island / Peninsula	23 (100%)	0 (0.0%)	0 (0.0%)	23 (100%)

$\chi^2 = 36.83$, $df = 4$, $p < .001$, Cramer's $V = 0.51$. Three of nine cells had an expected count below 5. Source: Field Survey, May 2026.

Table 5 shows that being unable to reach a facility because of transport problems was universal among island and peninsula respondents, all 23 of whom reported it, and in every case more than once. Most riverbank households (21 of 28, or 75.0 percent) had also had this experience, 15 of them more than once, whereas a majority of mainland households, 13 of 19 or 68.4 percent, reported it had never happened to them. The association is statistically significant, though small expected cell counts mean the exact chi-square value should be read with some caution.

Table 6: Settlement Type by Perceived Healthcare Access in Community

Settlement Type	Good	Poor	Very Poor	Total
Mainland	16 (84.2%)	3 (15.8%)	0 (0.0%)	19 (100%)
Riverbank	8 (28.6%)	14 (50.0%)	6 (21.4%)	28 (100%)
Island / Peninsula	0 (0.0%)	0 (0.0%)	23 (100%)	23 (100%)

$\chi^2 = 65.36$, $df = 4$, $p < .001$, Cramer's $V = 0.68$. One of nine cells had an expected count below 5. Source: Field Survey, May 2026.

The pattern in respondents' own assessment of healthcare access is stark. Every island and peninsula respondent rated access to major healthcare facilities as very poor, while 84.2 percent of mainland respondents rated it good and none rated it very poor. Riverbank respondents were spread across all three categories, consistent with their intermediate position on the objective travel time and cost measures reported earlier. This is the strongest evidence in the chapter that perceived access to care, one of the non-spatial factors named in the third

research objective, is itself shaped by settlement type rather than operating independently of it. It should be noted that this item measured respondents' assessment of access, not of clinical service quality; the two are distinct constructs and the survey instrument did not capture the latter.

Table 7: Settlement Type by Whether a Delay in Reaching Care Has Ever Harmed a Household Member's Health (Self-Reported)

Settlement Type	Yes, significantly	Somewhat	No	Don't know	Total
Mainland	3 (15.8%)	4 (21.1%)	12 (63.2%)	0 (0.0%)	19 (100%)
Riverbank	19 (67.9%)	3 (10.7%)	5 (17.9%)	1 (3.6%)	28 (100%)
Island / Peninsula	23 (100%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	23 (100%)

$\chi^2 = 35.59$, $df = 6$, $p < .001$, Cramer's $V = 0.50$. Seven of twelve cells had an expected count below 5; interpret with some caution. Source: Field Survey, May 2026.

Table 7 reports responses to the survey question "Has a delay in reaching care ever harmed a health outcome?" Every island and peninsula respondent answered yes, significantly, compared with 67.9 percent of riverbank respondents and 15.8 percent of mainland respondents. The association is significant, though more than half the cells in this table have small expected counts, so the precise chi-square value is best read as indicative rather than exact. Even allowing for that caveat, the direction and size of the gap between settlement types is unmistakable. The question referred to health outcomes in general terms. The survey did not record the specific condition or outcome respondents had in mind, and no outcome was verified against clinical records, so the finding measures perceived harm from delay and should not be read as a measure of morbidity or mortality.

Table 8: Settlement Type by Whether Household Has Unmet Health Needs Due to Access Barriers

Settlement Type	Yes	No	Total
Mainland	7 (36.8%)	12 (63.2%)	19 (100%)
Riverbank	22 (78.6%)	6 (21.4%)	28 (100%)
Island / Peninsula	23 (100%)	0 (0.0%)	23 (100%)

$\chi^2 = 22.18$, $df = 2$, $p < .001$, Cramer's V = 0.56. Source: Field Survey, May 2026.

Unmet health need follows the same gradient and holds up robustly under testing, with no small-cell concerns to qualify it. Every island and peninsula household reported at least one unmet health need in the past year, compared with 78.6 percent of riverbank households and 36.8 percent of mainland households.

Two further comparisons in this section did not reach statistical significance, and they are worth reporting precisely because they show that these barriers do not operate everywhere in the same way. Table 9 reports the first of these. Whether a household member had ever chosen not to seek needed care showed no significant association with settlement type ($\chi^2 = 4.98$, df = 2, p = .083, Cramer's V = 0.27). A majority of respondents in every settlement category reported this happening: 19 of 23 on the islands (82.6 percent), 25 of 28 along the riverbank (89.3 percent), and 12 of 19 on the mainland (63.2 percent). The mainland proportion is lower than the other two, but the difference is not large enough relative to the sample size to reach significance. Forgoing care altogether therefore appears to be a widespread experience across the whole district rather than one confined to the most geographically isolated communities. This is a substantively important null result rather than an absence of findings. Every other outcome examined in this chapter produced a sharp settlement-type gradient, yet the decision to go without needed care did not. It suggests that geography determines how hard it is to reach a major facility, but that the reasons households end up not seeking care at all are shared across the district and are not reducible to distance alone. Bypassing the nearest facility

in favour of a more distant one was reported by 33 of the 70 households (47.1 percent): 8 of 23 island and peninsula households (34.8 percent), 9 of 19 mainland households (47.4 percent), and 16 of 28 riverbank households (57.1 percent). Among those 33 households, referral by a health worker was the most common reason (20 households), followed by poor quality of care at the nearer facility (8), a preference for the doctor at the further facility (3), and drugs or supplies being out of stock (1). Bypassing therefore reflects the referral system as much as distrust of nearer facilities, although quality concerns account for a sizeable minority of cases.

Table 9: Settlement Type by Whether a Household Member Has Ever Chosen Not to Seek Needed Care

Settlement Type	Yes	No	Total
Mainland	12 (63.2%)	7 (36.8%)	19 (100%)
Riverbank	25 (89.3%)	3 (10.7%)	28 (100%)
Island / Peninsula	19 (82.6%)	4 (17.4%)	23 (100%)

$\chi^2 = 4.98$, df = 2, p = .083, Cramer's V = 0.27 (not significant). Two of six cells had an expected count below 5. Source: Field Survey, May 2026.

Table 10: Education Level by Perceived Healthcare Access

Education Level	Good	Poor	Very Poor	Total
No formal education	7 (16.7%)	11 (26.2%)	24 (57.1%)	42 (100%)
Primary school	13 (54.2%)	6 (25.0%)	5 (20.8%)	24 (100%)
JHS	3 (100.0%)	0 (0.0%)	0 (0.0%)	3 (100%)
Vocational	1 (100.0%)	0 (0.0%)	0 (0.0%)	1 (100%)

$\chi^2 = 19.26$, df = 6, p = .004, Cramer's V = 0.37. Six of twelve cells had an expected count below 5; interpret with some caution. Source: Field Survey, May 2026.

Table 10 shows a significant association between education level and perceived access, with respondents who had no formal education more likely to rate access very poor (57.1 percent) than those with primary education (20.8 percent). This should be read carefully, however, since respondents with no formal education in this sample are also more likely to live in the harder-to-reach riverbank and island communities. The relationship may partly reflect this geographic overlap rather than a direct effect of education on how access is perceived.

Table 11: NHIS Registration Status by Whether Household Member Chose Not to Seek Needed Care

NHIS Status	Yes	No	Total
Active membership	30 (76.9%)	9 (23.1%)	39 (100%)
Lapsed membership	23 (85.2%)	4 (14.8%)	27 (100%)
Never registered	3 (75.0%)	1 (25.0%)	4 (100%)

$\chi^2 = 0.75$, df = 2, p = .688, Cramer's V = 0.10 (not significant). Two of six cells had an expected count below 5. Source: Field Survey, May 2026.

NHIS registration status showed no statistically significant association with whether a household had forgone needed care: households with active membership (76.9 percent), lapsed membership (85.2 percent), and no membership (75.0 percent) forwent care at similarly high rates. Active members were no less likely to forgo care than the others, and the highest rate was among lapsed members. With only four never-registered households, comparisons involving that group rest on very few observations and should not be over-interpreted. Set alongside the strong, significant associations between settlement type and both transport cost and boat-crossing requirement reported in Section 4.3, this points to a clear policy implication. An insurance card does little to close the access gap in a district where reaching a major facility in the first place is the binding constraint.

Table 12: Summary of Chi-Square Tests of Association

Variables Tested	χ^2	df	p	Cramer's V
Settlement type × Travel time	79.32	4	< .001	0.75
Settlement type × Transport cost	43.16	2	< .001	0.79
Settlement type × Boat crossing required	61.67	2	< .001	0.94
Settlement type × Ever unable to reach	36.83	4	< .001	0.51
Settlement type × Perceived access	65.36	4	< .001	0.68
Settlement type × Delay harmed outcome	35.59	6	< .001	0.50
Settlement type × Unmet health needs	22.18	2	< .001	0.56
Settlement type × Forwent care	4.98	2	.083	0.27
Education level × Perceived access	19.26	6	.004	0.37
NHIS status × Forwent care	0.75	2	.688	0.10

Source: Field Survey, May 2026.

Table 13: Coping Strategies Used Instead of a Major Facility in the Past 12 Months (Multiple Response, N = 70)

Strategy	Frequency	Percentage (%)
Traditional healer or herbalist	54	77.1
Prayer or faith healer	46	65.7
Bought medicine from a chemist without prescription	11	15.7
Self-medicated at home	10	14.3
Consulted a CHPS community health worker only	5	7.1
None, always went to a major facility	4	5.7

Respondents could select more than one option, so percentages do not sum to 100. Source: Field Survey, May 2026.

Traditional healer or herbalist use, reported by 77.1 percent of respondents, and prayer or faith healing, reported by 65.7 percent, were common across the whole sample but noticeably more frequent among island and peninsula respondents: 21 of 23, or 91.3 percent, reported herbalist use, against 10 of 19, or 52.6 percent, on the mainland. These practices are better understood as rational responses to structural inaccessibility than as a simple cultural preference for alternative medicine. When a biomedical facility is a costly, uncertain boat journey away, a nearby, familiar service becomes the practical substitute, regardless of a household's insurance status.

4.5 GIS-Modelled Travel Time Analysis

We constructed a multi-modal network by merging the cleaned OpenStreetMap road network with manually digitised boat-crossing links. Service areas were then generated for thresholds of 30, 60 and 120 minutes from the three major facilities. In addition, we computed an origin-destination matrix from the 70 household GPS points to the facilities; we retained the minimum total cost for each origin as the modelled travel time to the nearest facility.

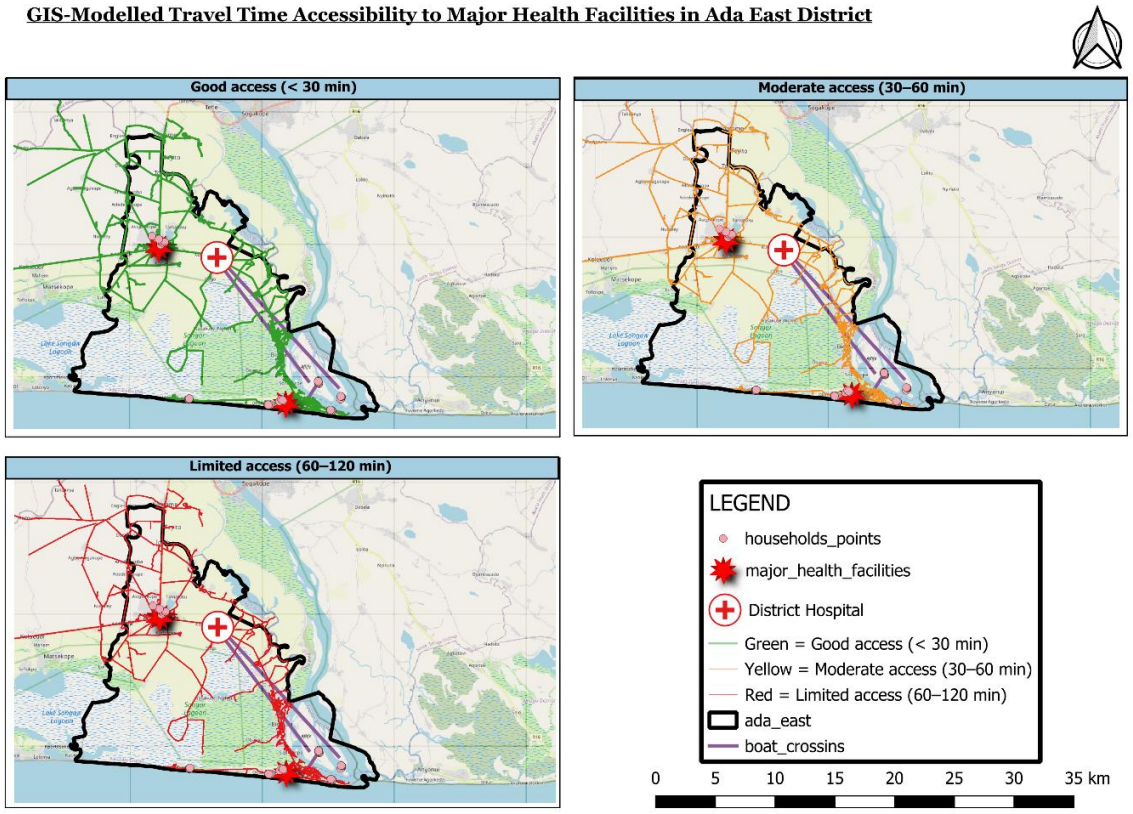


Figure 4: *GIS-Modelled Travel Time Accessibility to Major Health Facilities in Ada East District. Three-panel service area map showing good access (<30 min, green), moderate access (30–60 min, orange), and limited access (60–120 min, red), with household points, major facilities, and boat crossing links.*

Figure 4 shows the progressive expansion of the reachable network as the time threshold increases. The 30-minute service area (top left) is relatively restricted and concentrated around the major facilities and the principal road corridors north of the district.

The 60-minute service area (top right) covers a substantially larger portion of the road network, extending south toward the estuary communities. The 120-minute service area (bottom left) covers most of the remaining road network, but even at this generous threshold the reach does not extend across the water to the island and peninsula households clustered in the south-east of the district. The purple lines visible on all three panels indicate the digitised boat crossing links, which remain the only connection between the mainland road network and these communities.

When we extracted modelled times for individual household points, a clear geographic pattern emerged. Mainland households and many riverbank households returned realistic travel times, typically under 30 minutes for the closest facilities. In contrast, a substantial number of island and peninsula households returned null or zero values. This occurred because the network analysis algorithm could not identify a continuous path from those points to any major facility within the current topology of the multi-modal network. Although boat crossing lines had been digitised and assigned costs, they were not fully snapped and integrated at every landing point, or the household points themselves lay off the connected network. These null results are therefore best understood as a limitation of the model rather than as a measurement. They arise from incomplete topological integration of the digitised boat links, which is a correctable defect in the network dataset, and they cannot be read as independent evidence of the water barrier. What can be said is that the failure is systematic rather than random: every household for which the model failed was an island or peninsula household requiring a boat crossing, and no connected mainland or riverbank household failed. The direction of that failure is consistent with the structural discontinuity documented in the survey data, but the survey, not the network analysis, is what establishes it. Producing continuous modelled travel

times for these communities requires snapping the boat links to the road network at every landing point and re-running the origin-destination matrix, which is the immediate priority for revising this analysis.

Table 14: Comparison of GIS-Modelled and Self-Reported Travel Times for Surveyed Households (N = 70)

Self-Reported Category	Modelled: < 30 min	Modelled: 30–60 min	Modelled: No path (boat crossing gap)	Total
< 30 min	27	0	0	27
30–60 min	20	0	4	24
60–120 min	0	0	19	19
Total	47	0	23	70

Modelled returns of zero or null are reported together as 'No path' because both arise from the same cause, the absence of a traversable route in the network, rather than representing a travel time of zero. No household returned a modelled time in the 30–60 minute band, so that column is empty. Note that the categories in this table follow the classification in Section 3.3.3; the 60–120 minute band corresponds to the '1–2 hours' category used in Table 2. All 23 households that returned a null or zero modelled time were island or peninsula settlements that require boat crossing. These cases indicate that the network analysis could not identify a continuous route within the current multi-modal topology.

Source: Household GPS survey matched to QNEAT3 origin-destination matrix, May 2026.

Table 14 compares the GIS-modelled and self-reported travel times. For the subset of households that returned valid modelled times, agreement with self-reports was good for the shortest trips (both modelled and reported falling in the under-30-minute category). For intermediate trips, self-reported times were systematically higher than modelled times: modelled values of approximately 18–19 minutes still fell in the under-30-minute category, while respondents reported 30–60 minutes. This discrepancy is consistent with the fact that the pure network model does not fully capture boat waiting time, walking time to landing sites, or the unreliability of water transport. Households with null modelled times were predominantly those requiring boat crossings; their self-reported times fell mainly in the 60–120-minute category.

4.6 Household Voices: Priorities and Challenges in Their Own Words

The quantitative results presented above are grounded in the everyday experiences of the households surveyed. As part of the structured questionnaire, respondents were asked, in an open-ended item, what would most improve their access to healthcare and what challenges they currently face. A recurring theme was the compounded burden of distance, cost, and the boat crossing itself, which respondents described as inseparable from one another rather than as separate barriers. These responses were not coded or analysed thematically; they are reported to illustrate the quantitative patterns.

> *"Getting a boat to transport us in time and getting a doctor nearby to aid health care."* (Household Respondent, Ada East District)

For other respondents, the concern was less about reaching a facility at all and more about what happens once they arrive. Several described the quality and responsiveness of care itself as a barrier in its own right, distinct from the physical journey.

> *"Health workers are not responsive to emergency situations, and we also need a facility here."* (Household Respondent, Ada East District)
>
> *"Responses from doctor at facility is poor."* (Household Respondent, Ada East District)

This perception of unresponsiveness at public facilities appeared alongside an explicit preference for private care among some households, despite its higher cost, on the grounds of speed and attentiveness.

> *"Private is closer and attends faster. Everything is better with private."* (Household Respondent, Ada East District)

Fewer respondents raised communication as a barrier distinct from distance or cost, and several called for a facility or doctor to be based within their community rather than reached through improved transport alone.

> *"Language barrier at health facilities is a problem."* (Household Respondent, Ada East District)

> *"Doctors should be provided in the community, government should please intervene for us."* (Household Respondent, Ada East District)

Taken together, these responses point to concerns that extend beyond travel time and cost alone: the perceived responsiveness and quality of care once a facility is reached, and communication barriers that a purely spatial analysis cannot capture. The full set of responses to this item is retained in the researcher's field records.

4.7 Discussion

The findings are discussed in relation to the literature and the theoretical frameworks set out in Chapter Two. Taken together, the survey and GIS results support one central argument: settlement type, defined chiefly by whether a household must cross water, is the dominant determinant of accessibility to major healthcare facilities in Ada East District.

Distance and the water crossing. The self-reported data show a sharp divide. Not one island or peninsula respondent reported a travel time under 30 minutes, while 94.7 percent of mainland respondents did, and every island and peninsula respondent paid more than GHS 60 one way. The effect sizes are large (Cramer's V = 0.75 for travel time and 0.79 for transport cost). This is consistent with the well-established finding that distance and travel burden are among the strongest predictors of health facility use (Buor, 2003; Gething et al., 2012). What the Ada East evidence adds is that, where a water crossing is involved, the relevant measure of

distance is whether the household must cross water, not the number of kilometres. This echoes Boakye et al. (2026), who found that straight-line proximity can coexist with poor effective access, and it supports the argument in Chapter Two that road-only and proximity measures understate difficulty in water-fragmented districts (Neutens, 2015; Ray & Ebener, 2008). The GPS distances reinforce the point: island and peninsula households sit a median of only 1.8 km from the nearest major facility, yet face the longest, costliest journeys of any group (Section 4.3.4).

The second delay. In terms of the Three Delays Model (Thaddeus & Maine, 1994), these findings concern the second delay, reaching an appropriate facility. The survey shows that this delay has the compound structure anticipated in Section 2.1.2: for island and peninsula households it consists of a water segment and a road segment, each adding time, cost, and risk. Every island and peninsula household reported having been unable to reach a facility because of transport problems and having experienced harm from delay (Tables 5 and 7). Because these are self-reports and the specific conditions were not recorded, they show perceived consequences rather than measured clinical outcomes. The Three Delays Model was developed for maternal mortality, so the results suggest that the second delay is also salient for the wider range of time-sensitive needs households face, although this study cannot say which conditions were involved.

Enabling factors and their interaction. Andersen's model treats spatial accessibility, transport, and insurance as enabling factors and draws attention to how they interact (Andersen, 1968, 1995). The results fit that expectation. Settlement type, which determines travel burden, was strongly associated with transport cost, perceived access, delay-related harm, and unmet need (Cramer's V of 0.79, 0.68, 0.50, and 0.56), whereas NHIS status, a financial enabling

resource, was not significantly associated with forgoing care (Table 11). This is consistent with evidence that out-of-pocket costs remain a major source of hardship even where insurance nominally covers services (WHO & World Bank, 2021), since transport costs fall outside what NHIS covers. The association between education, a predisposing characteristic, and perceived access (Table 10) should be read cautiously, because respondents with no formal education were also more likely to live in the harder-to-reach settlements, so education and geography overlap. Need factors were not measured directly in this study.

Dimensions of access. In the terms of Penchansky and Thomas (1981), the survey measured accessibility (travel time), affordability (transport cost), and, through the perceived access item, an overall judgement of access. The comments reported in Section 4.6 point to dimensions the questionnaire did not measure directly: the responsiveness of staff and language barriers speak to acceptability and accommodation, and the calls for a doctor to be based in the community speak to availability. They also illustrate the point made by Levesque et al. (2013) that supply-side barriers and the abilities people need to use services compound one another, as they do here for households that are poor, have little formal education, and live across the water.

Forgoing care, bypassing, and coping. Forgoing needed care was the one outcome with no settlement-type gradient, occurring at high rates on the mainland as well as on the islands. Distance therefore appears to determine how hard the journey is, but not alone whether households make it. The literature offers a partial explanation: people bypass or avoid nearer facilities when they distrust the quality of care (Kruk et al., 2009), and respondents here described unresponsive staff and a preference for private care. The bypassing data add a qualification, however. Among the 33 households that had bypassed the nearest facility,

referral by a health worker was the most common reason (20 households), and poor quality of care (8), preference for a particular doctor (3), and stock-outs (1) accounted for fewer, so quality concerns explain only part of bypassing. The high use of traditional healers and faith healing, especially among island households, matches the feedback loop in the conceptual framework (Section 2.1.3), in which negative experiences of delay and cost reinforce reliance on informal care. It is better understood as a rational adaptation to structural inaccessibility than as a cultural preference. These interpretations remain tentative, because the survey was not designed to establish why households forgo care.

Modelled and self-reported travel time. For connected households, self-reported times exceeded modelled times, which agrees with validation studies finding that modelled estimates understate the times patients report (Dotse-Gborgbortsi et al., 2022; Rudolfson et al., 2020). The likely sources are waiting, walking to landing sites, and unreliable transport, which network speeds do not capture. The failure of the network model for every island and peninsula household illustrates in practice the limitation noted by Ray and Ebener (2008): standard network tools do not model boat transport without deliberate modification. As set out in Section 4.5, that failure reflects incomplete topology and cannot be treated as corroborating evidence of the water barrier. In this study the survey, not the GIS model, establishes the island and peninsula disadvantage.

Implications and limits. The evidence supports the conclusion that improving access to major healthcare facilities in Ada East District requires investment in the water-transport link, whether through more reliable boat services, shorter waiting times, or fixed crossings, in addition to any expansion of facility infrastructure or insurance coverage. These conclusions apply to the seven sampled communities rather than to the whole district, because respondents

were selected by convenience and were predominantly female (Section 3.3.2). Completing a topologically integrated multi-modal network remains a priority for future research so that continuous modelled travel times can be produced for every settlement.

CHAPTER FIVE: SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

This final chapter draws together the findings presented in Chapters One through Four. It summarises the study's results against the three research objectives set out in Section 1.3, states a conclusion in relation to the hypothesis set out in Section 1.4 and tested in Section 4.4, and offers recommendations for policy, practice, and further research.

5.2 Summary of Findings

The first objective was to map the spatial distribution of major health facilities in relation to surveyed settlements. This mapping (Section 4.1) showed a clear structural pattern: the Ada East District Hospital and Kasseh Health Centre sit together in the northern part of the district on the main road network, while Ada Health Centre lies apart from them, near the coast at Ada Foah. Surveyed households formed two corresponding clusters, a northern mainland cluster and a southern cluster of island, peninsula, and riverbank communities along the Volta estuary and the Songor Lagoon. This produces a north-south accessibility gradient that the subsequent household survey and hypothesis tests go on to quantify.

The second objective was to analyse self-reported travel time, transport mode, and transport cost by settlement type, and to complement these with GIS-modelled travel times. The self-reported results were unambiguous. Settlement type was very strongly associated with both travel time ($\chi^2 = 79.32$, df = 4, $p < .001$, Cramer's V = 0.75) and transport cost ($\chi^2 = 43.16$, df = 2, $p < .001$, Cramer's V = 0.79). Every island and peninsula respondent reported a journey of at least 30 minutes and a one-way cost above GHS 60, while most mainland respondents reported journeys under 30 minutes at lower cost. Boat crossing dependency was near-total for

island and peninsula households (Cramer's V = 0.94), though this figure is better read as confirmation that the settlement-type classification captured the intended geographic distinction than as an independent finding. The GIS-modelled half of this objective was only partially achieved: incomplete topological integration of the digitised boat crossing links meant the network model returned null travel times for the 23 island and peninsula households rather than usable estimates, so the planned comparison between modelled and self-reported travel time could only be completed for mainland and riverbank households (Section 4.5). This remains the study's most significant unresolved technical limitation.

The third objective was to investigate non-spatial factors, specifically perceived access, transport cost, and insurance coverage, and how they interact with settlement type. Perceived access, perceived delay-related harm to health, and current unmet health need were all significantly associated with settlement type, with effect sizes (Cramer's V of 0.68, 0.50, and 0.56 respectively) approaching those found for the objective travel time and cost measures. NHIS registration status, by contrast, showed no significant association with whether a household had ever forgone needed care (Cramer's V = 0.10), and households with active, lapsed, and no membership forwent care at similarly high rates (76.9, 85.2, and 75.0 percent respectively). Forgoing care was also the one outcome in the entire study that showed no settlement-type gradient: mainland, riverbank, and island households forwent care at broadly similar, and uniformly high, rates. This is arguably the most important single finding of the study: distance and settlement type determine how hard the journey to care is, but the decision to forgo that journey altogether appears to be driven by factors common across the whole district rather than by geography alone.

5.3 Conclusions

In relation to the study's hypothesis, the null hypothesis (H_0), that there is no statistically significant relationship between settlement type and spatial accessibility to major healthcare facilities, is rejected. The alternative hypothesis (H_1) is supported: settlement type was very strongly and significantly associated with self-reported travel time ($\chi^2 = 79.32$, $df = 4$, $p < .001$, Cramer's V = 0.75), transport cost ($\chi^2 = 43.16$, $df = 2$, $p < .001$, Cramer's V = 0.79), and perceived healthcare access ($\chi^2 = 65.36$, $df = 4$, $p < .001$, Cramer's V = 0.68), with island and peninsula communities faring significantly worse than mainland and riverbank communities on every measure.

One finding fell outside this pattern and is reported as an additional, exploratory result rather than as part of the formal hypothesis test. NHIS registration status showed no statistically significant association with whether a household had ever forgone needed care ($\chi^2 = 0.75$, $df = 2$, $p = .688$, Cramer's V = 0.10); households with active membership forwent care at a rate (76.9 percent) similar to that of lapsed (85.2 percent) and never-registered (75.0 percent) households. This suggests that financial protection through NHIS membership does not, on its own, compensate for the structural barrier created by physical geography.

Taken together, these conclusions support the central argument of the study: physical geography, and specifically the presence or absence of a water crossing, is the dominant driver of healthcare accessibility difficulty in Ada East District. It shapes not only objective measures of travel time and cost but also perceived access and the risk of care-related harm. Financial protection through NHIS membership, while valuable, does not on its own compensate for this structural barrier, and the decision to forgo care altogether is better explained by factors shared across the district than by settlement type. The study also concludes, on methodological grounds, that self-reported survey data remains the more reliable source of accessibility

evidence for island and peninsula communities in this district until the underlying multi-modal network model is fully topologically resolved.

Contribution to knowledge. The literature reviewed in Chapter Two established that distance, cost, and perceived quality shape healthcare use, but it had not shown how these operate where a water crossing, rather than road distance, is the main barrier to major facilities. This study adds three things. First, it shows that in the sampled Ada East communities the presence of a water crossing separates households' travel time, transport cost, perceived access, and perceived harm from delay very sharply. Second, it shows that NHIS registration did not distinguish households' likelihood of forgoing care, which is consistent with the view that financial coverage does not offset the geographic barrier. Third, it provides methodological evidence that a multi-modal network model with incompletely integrated boat links fails systematically for exactly the households most affected, so that survey evidence remains the more reliable measure for these communities until the network is completed.

5.4 Recommendations

Based on these findings, the following recommendations are made.

First, the Ghana Health Service and the Ada East District Health Directorate should treat water transport, rather than facility construction alone, as a priority intervention for island and peninsula communities. A subsidised or scheduled boat ambulance service linking these communities to the Ada East District Hospital and Kasseh Health Centre would directly address the transport cost and delay barriers identified in Sections 4.3 and 4.4, which current facility-based planning does not reach.

Second, the District Health Directorate should formally review the staffing classification of Ada Foah Health Centre. As noted in Sections 1.2 and 1.6, the facility is

currently led by a Physician Assistant rather than a resident medical doctor, yet it remains the only in-patient and emergency-capable facility serving the Ada Foah area. Clarifying whether this facility is resourced and staffed appropriately for the volume and severity of cases it serves would strengthen both service delivery and future research that relies on it as a destination facility.

Third, the National Health Insurance Authority and the District Health Directorate should investigate why NHIS membership does not reduce the likelihood of forgoing care in this district. Since the barrier does not appear to be financial coverage alone, factors such as the availability of NHIS-accredited services at accessible facilities, indirect costs not covered by the scheme (transport, lost income, informal payments), and confidence in the quality of covered care warrant direct investigation, potentially through the key informant interviews planned but not yet completed for this study.

Fourth, the Ada East District Assembly, in collaboration with the Ghana Highways Authority and relevant development partners, should treat the water crossings identified in this study (Section 4.1, Figure 3) as candidate sites for future infrastructure investment, whether through improved landing points, subsidised ferry services, or, where feasible, fixed crossings. Even where full network modelling of these routes remains incomplete, the pattern established through the household survey is sufficiently strong to justify prioritisation.

Fifth, agencies conducting future geospatial accessibility studies in riverine or island districts should budget explicitly for the topological integration of water transport links into road network models before analysis begins. This study's own experience, in which incomplete snapping of boat crossing links produced unusable null values for a third of the sample, demonstrates that this step cannot be treated as a minor technical detail.

5.5 Suggestions for Further Research

Several directions for further research follow directly from this study's limitations, discussed in Section 1.6.

The multi-modal GIS network constructed for this study should be completed by snapping the digitised boat crossing links to the road network at every landing point and re-running the origin-destination matrix in QNEAT3. This would allow the comparison between modelled and self-reported travel time, planned as part of the second objective, to be completed for the full sample rather than the mainland and riverbank subset.

The key informant interviews with health workers and community leaders, for which ethical clearance was still pending at the time of this study, should be completed. These would help explain the district-wide, settlement-independent pattern of forgoing care identified in Section 4.4, which the household survey alone cannot fully account for.

Future studies should consider a probability-based sampling design capable of supporting district-wide statistical inference, since the convenience sample used here, while adequate for identifying patterns within the seven sampled communities, cannot be generalised further.

Given that women accounted for 84.3 percent of respondents in this study, a result of deliberately targeting household caretakers, future research should consider designs that capture male household members more evenly, to test whether the accessibility patterns identified here hold consistently across genders.

Finally, further research should investigate directly why the decision to forgo needed care shows no settlement-type gradient in this district, since this was the study's most unexpected finding and is not explained by any of the spatial or insurance-related variables examined here.

REFERENCES

Andersen, R. M. (1968). *A behavioral model of families' use of health services*. Center for Health Administration Studies, University of Chicago.

Andersen, R. M. (1995). Revisiting the behavioral model and access to medical care: Does it matter? *Journal of Health and Social Behavior, 36*(1), 1–10.

Boakye, P. A., Buor, D., & Tei-Nartey, F. (2026). Geospatial analysis of primary healthcare accessibility and qualitative assessment of affordability in rural communities of Twifo Ati-Morkwaa District, Ghana. *BMC Primary Care*. https://doi.org/10.1186/s12875-026-03260-2

Buor, D. (2003). Analysing the primacy of distance in the utilization of health services in the Ahafo-Ano South district, Ghana. *International Journal of Health Planning and Management, 18*(4), 293–311.

Dotse-Gborgbortsi, W., Nilsen, K., Ofosu, A., Matthews, Z., Tejedor-Garavito, N., Wright, J., & Tatem, A. J. (2022). Distance is "a big problem": A geographic analysis of reported and modelled proximity to maternal health services in Ghana. *BMC Pregnancy and Childbirth, 22*, Article 672.

Gething, P. W., Johnson, F. A., Frempong-Ainguah, F., Nyarko, P., Baschieri, A., Aboagye, P., Falkingham, J., Matthews, Z., & Atkinson, P. M. (2012). Geographical access to care at birth in Ghana: A barrier to safe motherhood. *BMC Public Health, 12*, Article 991.

Ghana Statistical Service. (2021). *2021 population and housing census, volume 2: Proximity of residential structures to essential service facilities*.

Higgs, G., Langford, M., Jarvis, P., Page, N., Richards, J., & Fry, R. (2019). Using geographic information systems to investigate variations in accessibility to "extended hours"

primary healthcare provision. *Health & Social Care in the Community, 27*(4), 1074–1084.

Kruk, M. E., Mbaruku, G., McCord, C. W., Moran, M., Rockers, P. C., & Galea, S. (2009). Bypassing primary care facilities for childbirth: A population-based study in rural Tanzania. *Health Policy and Planning, 24*(4), 279–288.

Levesque, J.-F., Harris, M. F., & Russell, G. (2013). Patient-centred access to health care: Conceptualising access at the interface of health systems and populations. *International Journal for Equity in Health, 12*, Article 18.

Luo, W., & Wang, F. (2003). Measures of spatial accessibility to health care in a GIS environment: Synthesis and a case study in the Chicago region. *Environment and Planning B: Planning and Design, 30*(6), 865–884.

Neutens, T. (2015). Accessibility, equity and health care: Review and research directions for transport geographers. *Journal of Transport Geography, 43*, 14–27.

Okafor, F. C. (1990). The spatial dimensions of accessibility to general hospitals in rural Nigeria. *Socio-Economic Planning Sciences, 24*(4), 295–306.

Penchansky, R., & Thomas, J. W. (1981). The concept of access: Definition and relationship to consumer satisfaction. *Medical Care, 19*(2), 127–140.

Ray, N., & Ebener, S. (2008). AccessMod 3.0: Computing geographic coverage and accessibility to health care services using anisotropic movement of patients. *International Journal of Health Geographics, 7*, Article 63.

Rudolfson, N., Gruendl, M., Nkurunziza, T., Nyirasafari, R., Nkeshimana, M., Kumar, P., Riviello, R., & Citron, I. (2020). Validating the global surgery geographical

accessibility indicator: Differences in modeled versus patient-reported travel times. *World Journal of Surgery, 44*, 2123–2130.

Thaddeus, S., & Maine, D. (1994). Too far to walk: Maternal mortality in context. *Social Science & Medicine, 38*(8), 1091–1110.

World Health Organization. (2010). *Monitoring the building blocks of health systems: A handbook of indicators and their measurement strategies*.

World Health Organization. (2018). *Primary health care: Transforming and sustaining health*.

World Health Organization & World Bank. (2021). *Tracking universal health coverage: 2021 global monitoring report*.

APPENDIX A: FIELD PHOTOGRAPHS

The photographs below were taken during the May 2026 fieldwork period and illustrate the study area, the settlements surveyed, and the water crossings central to this study.



Plate 1: *The researcher with respondents at Kasseh during household data collection.*



Plate 2: *The researcher with respondents at a mainland household who welcomed the survey team.*



Plate 3: *Canoes moored at the Ada Foah crossing point used to reach the island and peninsula communities.*



Plate 4: *The researcher wearing a life jacket while crossing by boat to the island communities.*



Plate 5: *A coastal view near Azizanya, showing fishing activity typical of the riverbank and peninsula communities surveyed.*



Plate 6: *The researcher with a respondent and a translator during an interview at Azizanya.*

APPENDIX B: HOUSEHOLD SURVEY QUESTIONNAIRE

INSTRUMENT A: HOUSEHOLD QUESTIONNAIRE

Spatial Accessibility to Major Healthcare Facilities in Ada East District. University of Ghana, Department of Geography and Resource Development.

The questionnaire was administered in the KoboCollect mobile application. It is a condensed version of the paper Instrument A designed for the study; the items below are those that appear in the KoboCollect form and the dataset.

Interview details

Who are you interviewing? Household respondent (Instrument A)

Enumerator name; Date; Community / Settlement

Settlement type: ☐ Mainland ☐ Island / Peninsula ☐ Riverbank

GPS coordinates (recorded at the respondent's household)

Informed consent

I am a University of Ghana student researcher studying healthcare access in Ada East District. Participation is voluntary and your answers will be kept confidential. The interview takes about 15 minutes. You may stop at any time.

Do you consent to participate? ☐ Yes (proceed) ☐ No (thank respondent and end interview)

Section A: Demographic information

A1. Sex. ☐ Male ☐ Female ☐ Other

A2. Age group. ☐ 18–25 ☐ 26–35 ☐ 36–45 ☐ 46–55 ☐ 56–65 ☐ Over 65

A3. Highest education level. ☐ No formal education ☐ Primary (Basic 1–6) ☐ Junior High School (JHS) ☐ Senior High School (SHS) ☐ Tertiary / University ☐ Vocational / Technical ☐ Other

A4. Primary occupation. ☐ Fishing ☐ Farming / Agriculture ☐ Salt harvesting ☐ Trading / Petty commerce ☐ Government / Public sector worker ☐ Self-employed / Artisan ☐ Student ☐ Unemployed / Homemaker ☐ Other

A5. Household size. ☐ 3–5 people ☐ 6–9 people ☐ 10 or more

A6. Registered with the National Health Insurance Scheme (NHIS)? ☐ Yes, active membership ☐ Yes, but expired ☐ No, never registered

A7. If not a member, what is the main reason? (Skip if active member)

Section B: Location, transport, and travel to major facilities

B1. Main mode of transport to the district hospital or nearest major facility? ☐ Walk (footpath only) ☐ Walk then public transport (trotro / bus) ☐ Motorbike / Okada ☐ Private car / taxi ☐ Boat / canoe only ☐ Boat then road transport ☐ Ambulance (emergency) ☐ Other

B2. How long does it take to reach the nearest major facility? ☐ Less than 30 min ☐ 30–60 min ☐ 1–2 hours ☐ 2–3 hours ☐ More than 3 hours

B3. Typical one-way transport cost (GHS)? ☐ Free (walk) ☐ GHS 1–10 ☐ GHS 11–30 ☐ GHS 31–60 ☐ More than GHS 60

B4. Does the journey require a boat or canoe crossing? ☐ Yes ☐ No

B5. Have you ever been unable to reach a facility because of transport problems? ☐ Yes, more than once ☐ Yes, once ☐ No

B6. If yes, what caused the problem? (Tick all that apply) ☐ No boat / canoe available ☐ Bad weather / strong currents ☐ Could not afford transport fare ☐ No vehicle / motorbike available ☐ Road was flooded / impassable ☐ No one to accompany me ☐ Had to leave very early in the morning ☐ Other

B7. When is it hardest to reach a major facility? ☐ Rainy season (April–July) ☐ Second rains (Sept–Nov) ☐ Dry season (Dec–March) ☐ No difference by season ☐ I do not know

Section C: Healthcare-seeking behaviour and facility use

C1. In the past 12 months, did any household member visit a major facility? ☐ Yes ☐ No

C2. If yes, which facility did they visit most recently? ☐ Ada Foah Government Hospital ☐ Private clinic ☐ Other

C3. Why that particular facility? (Tick all that apply) ☐ It is the nearest major facility ☐ Better doctors / equipment ☐ Referred there ☐ NHIS coverage accepted here ☐ Recommended by others ☐ Had a bad experience elsewhere ☐ Only facility that can treat my condition ☐ Other

C4. Did any household member choose NOT to seek care despite needing it? ☐ Yes ☐ No ☐ Don't know

C5. If yes, why not? (Tick all that apply) ☐ Too far / too long to travel ☐ Could not afford transport ☐ Could not afford treatment ☐ Condition improved on its own ☐ Visited a traditional healer instead ☐ No one to accompany them ☐ Facility perceived as poor quality ☐ Fear of hospitals / medical staff ☐ Family / cultural reasons ☐ Drugs / supplies usually out of stock ☐ Staff are not helpful ☐ It is closed / no staff available ☐ Other

C6. Has anyone in the household bypassed the nearest facility to go further? ☐ Yes ☐ No

C7. If yes, main reason for bypassing? ☐ Poor quality of care at nearer facility ☐ Drugs / supplies usually out of stock ☐ Staff are unfriendly or incompetent ☐ Prefer the doctor at the further facility ☐ Referred by health worker ☐ Other

C8. Has a delay in reaching care ever harmed a health outcome? ☐ Yes, significantly ☐ Somewhat ☐ No ☐ Don't know

Section D: Perceptions of healthcare access and quality

D1. How would you describe healthcare access in your community? ☐ Very good ☐ Good ☐ Poor ☐ Very poor ☐ Don't know

D2. Level of agreement with each statement (Strongly agree / Agree / Neutral / Disagree / Strongly disagree): (a) The nearest major facility is easy for me to reach. (b) Transport cost to the facility is affordable. (c) I can always find transport when I need care. (d) I trust the quality of care at the facility I use. (e) The facility usually has the medicines and equipment I need.

D3. What is the biggest barrier to accessing healthcare here? (Open response)

Section E: Unmet needs and recommendations

E1. Are there health needs unmet in your household due to access barriers? ☐ Yes ☐ No

E2. In the past 12 months, has a household member used any of the following instead of a major facility? (Tick all that apply) ☐ Traditional healer / herbalist ☐ Prayer / faith healer ☐ Bought medicine from a chemist without prescription ☐ Self-medicated with drugs at home ☐ Consulted a CHPS community health worker only ☐ None — always went to a major facility

E3. What health service do you most urgently need closer to home? ☐ Emergency / accident care ☐ Maternal / childbirth services ☐ Child health and vaccination ☐ Surgery ☐ Laboratory / diagnostic services ☐ Mental health services ☐ Chronic disease management (diabetes, hypertension) ☐ Eye / dental care ☐ Other

E4. Any other comments? (Open response)